

# Yuliang Peng

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## EDUCATION

### University of Maryland, College Park

- Bachelor of Science in Computer Science (Dean's List) **Graduated : May 2024**
- Master of Science in Machine Learning **Expected Graduation: May 2027**
- Relevant Courses: Machine Learning, Computer Vision, Robotics, Deep Learning, Data Structures and Algorithms

## WORK EXPERIENCE / RESEARCH EXPERIENCE

**University of Maryland, College Park** *Research Assistant* **Maryland, USA** Nov 2025 - Present

- Developed GPU-accelerated spatiotemporal embeddings for high-rate event camera streams, enabling efficient ML inference with optimized CUDA-based parallelization.
- Conducted data-driven analysis of aviation network resilience using network-level modeling on Delta system outage data.
- Applied NLP and sentiment analysis to TripAdvisor low-altitude travel reviews to extract passenger perception dimensions

**ModeSens** *Software Engineer Full-Time* **Remote, USA** Jun 2025 - Sep 2025

- Built a distributed e-commerce crawler using asynchronous pipelines and serverless triggers to automatically ingest product data from 30+ merchant catalogs with AI-based analysis.
- Built an NLP pipeline to normalize user search queries against merchant catalogs, improving search precision by 13%..
- Implemented an image-based product matching workflow that compares user-uploaded images against internal product catalogs to support visual search and recommendations.

**Bank of China, USA** *Software Engineer Full-Time* **Los Angeles, USA** July 2024 - May 2025

- Led bug bash and reliability efforts for internal enterprise systems, resolving 70+ high-impact issues and reducing productivity-blocking tickets by 10%.
- Implemented a secure client authorization framework using cryptographic signatures and the Chain of Responsibility pattern, adopted by 15+ internal teams.
- Built AWS-based data pipelines (EC2, S3, RDS) for real-time ingestion and storage of financial transaction data, improving data reliability and operational efficiency.

**Air China** *Software Engineer Intern* **Sichuan, China** May 2023 - August 2023

- Developed Unix shell scripts to automate system backups across Linux environments, reducing manual workload by 40%.
- Revamped the secure login system with mobile QR code scanning, streamlining user identification. (adopted in 2024)

## PROJECTS

**Tabit – AI-Powered Browser Context Manager | Google Chrome Extension Hackathon | [Github Code](#)**

- Built an **AI-powered Chrome extension** (React, Vite, Manifest V3) to automatically cluster and organize browsing tabs.
- Implemented **similarity-based tab grouping** with hierarchical clustering and Web Worker parallel processing to manage **100+ tabs in real time**.
- Designed a **session memory system** to persist and restore tab groups across browsing sessions.

**FaceID-Style Verification System | LFW Dataset | [Github Code](#)**

- Built an end-to-end **face verification system** on the LFW dataset using similarity-based embedding matching.
- Designed a **reproducible ML pipeline** for data ingestion and pair generation with ROC-based threshold calibration.
- Packaged the inference workflow using **CLI interface** and benchmarked latency under concurrent workloads.

**Image-Based Product Matching Tool | Machine Learning | [Github Code](#)**

- Built an image similarity system using perceptual hashing and SSIM to match product images.
- Developed a **Python-based image extraction pipeline** with Selenium, reducing manual matching workload by **80%** and improving recommendation accuracy by **25%**.

## SKILLS

**Programming Languages:** C++, CUDA, Python, C, Java, Dart, JavaScript, Unix Shell, SQL

**Tools / Frameworks:** AWS, Docker, Redis, NumPy, Pandas, Flask, Spring Boot, Vue.js, Flutter, Git, Sentry